

DRAFT C++ Synchronized Buffered Ostream

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Introduction

At present, some stream output operations guarantee that they will not produce race conditions, but do not guarantee that the effect will be sensible. Some form of external synchronization is required. Unfortunately, without a standard mechanism for synchronizing, independently developed software will be unable to synchronize.

[N3535 C++ Stream Mutexes](#) proposed a standard mechanism for finding and sharing a mutex on streams. At the Spring 2013 standards meeting, the Concurrency Study Group requested a change away from a full mutex definition to a definition that also enabled buffering.

[N3678 C++ Stream Guards](#) proposed a standard mechanism for batching operations on a stream. That batching may be implemented as mutexes, as buffering, or some combination of both. It was the response to the Concurrency Study Group. A draft of that paper was reviewed in the Library Working Group, who found too many open issues on what was reasonably exposed to the 'buffering' part.

[N3665 Uninterleaved String Output Streaming](#) proposed making streaming of strings of length less than `BUFSIZ` appear uninterleaved in the output. It was a "minimal" functionality change to the existing standard to address the problem. The full Committee chose not to adopt that minimal solution.

[N3750 C++ Ostream Buffers](#) proposed an explicit buffering, at the direction of the general consensus in the July 2013 meeting of the Concurrency Study Group. In November 2014 a further updated version [N4187](#) was discussed in Library Evolution in Urbana and it was consensus to work with a library expert to get the naming and wording better suited to LWG expectations. Nico Josuttis volunteered to help the original authors. More information on the discussion is available at [LEWG wiki](#) and the corresponding [LEWG bug tracker entry \(20\)](#). This paper address issues raised with [N4187](#). This paper has a change of title to reflect a change in class name, but contains the same fundamental approach.

Solution

We propose a `basic_osyncstream`, that buffers output operations for a wrapped stream. The `basic_osyncstream` will atomically transfer the contents of an internal stream buffer to a `basic_ostream`'s stream buffer on destruction of the `basic_osyncstream`.

The transfer on destruction simplifies the code and ensures at least some output in the presence of an exception.

The intent is that the `basic_osyncstream` is an automatic-duration variable with a relatively small scope which constructs the text to appear uninterleaved. For example,

```
{
    ostream cout(cout);
    cout << "Hello, " << "World!" << '\n';
}
```

Design

Here we list a set of objection to our proposal in the form of questions. We then give our reasons why other potential solutions do not work as well as our proposed solution.

Why can I not just use cout? It should be thread-safe.

You will not get data races for cout, but that is not true for most other streams. In addition there is no guarantee that output from different threads appears in any sensible order. Coherent output order is the main reason for this proposal.

Why must ostream be an ostream? Could a simple proxy wrapper work?

To support all existing and user-defined output operators, ostream must be an ostream subclass.

Can you make a flush of the ostream mean transfer the characters and flush the underlying stream?

No, because the point of the ostream is to collect text into an atomic unit. Flushes are often emitted in calls where the body is not visible, and hence unintentionally break up the text. Furthermore, there may be more than one flush within the lifetime of an ostream, which would impose a performance loss when an atomic unit of text needs only one flush. The design decision is only to flush the underlying stream, if the ostream was flushed, and only once per atomic transfer of the character sequence.

Can you flush just transfer the characters and not flush the underlying stream?

The flush intends an effect on visible to the user. So, we should preserve at least one flush. The flush may not be visible to the code around the ostream, and so its programmer cannot do a manual flush, because attempting to flush the underlying stream that is shared among threads will introduce a data race.

While it has been pointed out that the implementation should actually reside in the used stream buffer object, we believe the specification should hide that effect, because such a stream buffer is not intended to be ever be used standalone. Therefore, the specification here only provides the basic_ostream class template and not any details on the underlying stream buffer implementing the mechanics. We have an implementation, where the basic_ostream is more or less just a shell to the real meat in the stream buffer but that stream buffer is not meant to be exchanged or shared or anything.

We follow typical stream conventions of basic_ prefixes and typedefs.

The constructor for ostream takes a non-const reference to a basic_ostream obtaining its stream buffer or a basic_streambuf. This stream buffer indicates that the destruction of the ostream may write to the stream buffer obtained with the constructor argument.

The wording below permits implementation of basic_ostream with either a stream_mutex from [N3535](#) or with implementations suitable for [N3665](#), e.g. with Posix file locks [\[PSL\]](#)

Feature Test

No header called <ostream> exists; testing for this header's existence is thus sufficient for testing existence of the feature.

Wording

This wording is relative to N4594.

17.6.1.2 Headers [headers]

Add a new entry to table 14 — C++ library headers :

<ostream>

27.1 General [input.output.general]

Add a new row to table 129 — Input/output library summary.

27.12 Synchronized output streams <syncstream>

27.3 Forward declarations [iostream.forward]

Add the following forward declarations to the synopsis of <iosfwd> in the namespace `std`.

```
template <class charT,  
         class traits = char_traits<charT>,  
         class Allocator = allocator<charT> >  
class basic_ostream;  
typedef basic_ostream<char> ostream;  
typedef basic_ostream<wchar_t> wostream;
```

27.12 Synchronized output stream [syncstream]

Add a new section with the following subsections.

27.12.1 Overview [syncstream.overview]

The header <syncstream> provides a mechanism to synchronize execution agents writing to the same stream. It defines a class template `basic_ostream` to buffer output and transfer the buffered content into an object of type `basic_streambuf<charT,traits>` atomically with respect to such transfers by other `basic_ostream<charT,traits,Allocator>` objects referring to the same `basic_streambuf<charT,traits>` object. The transfer occurs when `emit()` is called and when the `basic_ostream<charT,traits,Allocator>` object is destroyed.

Add a synopsis for header <syncstream>.

```
template <class charT,  
         class traits,  
         class Allocator>  
class basic_ostream;
```

27.12.2 Class template `basic_ostream` [syncstream.ostream]

```
template <class charT,  
         class traits,  
         class Allocator>  
class basic_ostream  
: public basic_ostream<charT,traits>  
{  
public:  
    typedef charT char_type;  
    typedef traits traits_type;  
    typedef typename traits::int_type int_type;  
    typedef typename traits::pos_type pos_type;  
    typedef typename traits::off_type off_type;  
    typedef Allocator allocator_type;  
    typedef basic_streambuf<charT,traits> streambuf_type;  
  
    explicit basic_ostream(streambuf_type *obuf,  
                           Allocator const &allocator = Allocator());  
    explicit basic_ostream(basic_ostream<charT,traits> &os,  
                           Allocator const &allocator = Allocator());  
    basic_ostream(basic_ostream&&);  
    ~basic_ostream();  
    basic_ostream& operator=(basic_ostream&&);  
    void emit();  
    streambuf_type* rdbuf() const noexcept;  
private:  
    streambuf_type *out; // exposition only  
};
```

Allocator shall meet the allocator requirements [allocator.requirements].

[Example: Use a named variable within a block statement for streaming across multiple statements.

```
{  
    ostream bout(cout);  
    bout << "Hello, ";  
    bout << "World!";  
    bout << endl; // flush is noted  
    bout << "and more!\n";  
} // characters are transferred and cout is flushed
```

—end example]

[Example: Use a temporary object for streaming within a single statement. cout is not flushed.

```
ostream(cout) << "Hello, " << "World!" << '\n';
```

—end example]

27.12.2.1 Constructor [syncstream.ostream.ctor]

```
explicit basic_ostream(streambuf_type *buf, Allocator const &allocator = Allocator());
```

Effects: Stores the stream buffer pointer buf which will be the final destination of characters in out. If the stored stream buffer pointer out == nullptr output operations on *this fail. A copy of allocator is used to allocate memory for the stream's own internal buffer, if any.

Constructs a stream buffer object [stream.buffer].

[Note: If wrapped stream buffer pointer refers to a user provided stream buffer then its implementation must be aware that its member functions might be called from emit() while a lock is held. —end note]

Postconditions: out == buf

```
explicit basic_ostream(basic_ostream<charT,traits>& os, Allocator const &allocator=Allocator());
```

Effects: basic_ostream(os.rdbuf(),a)

```
basic_ostream(basic_ostream&& os);
```

Effects: First, the state necessary to perform os.emit() is moved to *this. Then os.out = nullptr. [Note: This disassociates os from its wrapped stream buffer ensuring destruction of os produces no output. —end note]

Postconditions: The value returned by this->rdbuf_wrapped() is the value returned by os.rdbuf_wrapped() prior to calling this constructor. Output stored in *os.rdbuf() prior to calling this constructor will be stored in *this->rdbuf() afterwards. Then os.rdbuf()->pbase() == os.rdbuf()->pptr() and nullptr == os.rdbuf_wrapped().

27.12.2.2 Destructor [syncstream.ostream.dtor]

```
~basic_ostream();
```

Effects: Calls emit().

27.12.2.3 Assignment [syncstream.ostream.assign]

```
basic_ostream& operator=(basic_ostream&& os);
```

Effects: First, calls this->emit(). Then the state necessary to perform os.emit() is moved to *this. Then os.out = nullptr. [Note: This disassociates os from its wrapped stream buffer ensuring destruction of os produces no output. —end note]

Postcondition: Primarily, `nullptr == os.rdbuf_wrapped()`. Also, `rdbuf_wrapped()` returns the value previously returned by `os.rdbuf_wrapped()`.

27.12.2.4 Member Functions [syncstream.osyncstream.mfun]

`void emit();`

Effects: Transfers the contents of the internal stream buffer to the stream buffer `*out`, so that they appear in the output stream as a contiguous sequence of characters. If and only if a flush was requested on `*this`, `out->pubsync()` is called on the stream buffer obtained in the constructor. A failure in transferring the characters or flushing calls, such as `out` having the value `nullptr`, result in `setstate(ios::badbit)`.

[Note: A flush on a `basic_osyncstream` does not flush the `basic_osyncstream` to the stream buffer, but rather flushes the stream buffer after transferring the contents of the `basic_osyncstream` to the stream buffer.

[Example:

```
{
    osyncstream bout(cout);
    bout << "Hello," << '\n'; // no flush
    bout.emit(); // characters transferred; cout not flushed
    bout << "World!" << endl; // flush noted; bout not flushed
    bout.emit(); // characters transferred; cout flushed
    bout << "Greetings." << '\n'; // no flush
} // characters transferred; cout not flushed
```

—end example] —end note]

Synchronization: All `emit()` calls transferring characters to the same stream buffer object appear to execute in a total order consistent with *happens-before* where each `emit()` call *synchronizes-with* subsequent `emit()` calls in that total order.

Remarks: May hold a lock uniquely associated with the underlying stream buffer object obtained in the constructor while transferring characters.

[Example: The function `emit()` can be used to catch exceptions from operations on the underlying stream.

```
{
    osyncstream bout(cout);
    bout << "Hello, " << "World!" << '\n';
    try {
        bout.emit();
    } catch ( ... ) {
        // stuff
    }
}
```

—end example]

`streambuf_type* rdbuf_wrapped() const noexcept;`

Returns: `out`

[Example: Obtaining the wrapped stream buffer with `rdbuf_wrapped()` allows wrapping it again with an `osyncstream`. For example,

```
{
    osyncstream bout1(cout);
    bout1 << "Hello, ";
    {
        osyncstream bout2(bout1.rdbuf_wrapped());
        bout2 << "Goodbye, " << "Planet!" << '\n';
    }
    bout1 << "World!" << '\n';
}
```

produces the *uninterleaved* output

```
Goodbye, Planet!  
Hello, World!
```

—end example.]

Implementation

Two example implementations are available on

github.com/PeterSommerlad/SC22WG21_Papers/workspace/Test_basic_osyncstream and
https://github.com/PeterSommerlad/SC22WG21_Papers/tree/master/workspace/p0053_basic_osyncstreambuf.

The latter actually implements almost all of the mechanics in a stream buffer as proposed by Pablo Halpern where the first one is my original implementation with most of the mechanics stuck in the `basic_osyncstream`.

Revisions

This paper revises [P0053R2 C++ Synchronized Buffered Ostream](#)

- Remove the "may construct a mutex" notes.
- Remove "may destroy mutex" notes.
- Clarify `osyncstream` flush behavior in an example.
- Make minor editorial fixes.

[P0053R2](#) revised [P0053R1 C++ Synchronized Buffered Ostream](#)

- Provide a typedef for the wrapped stream buffer and use it to shorten the specification as suggested by Daniel Krügler.
- Provide move construction and move assignment and specify the moved-from state to be detached from the wrapped stream buffer.
- Rename `get()` to `rdbuf_wrapped()` and provide `noexcept` specification.
- Changed to explicitly rely on wrapping a stream buffer, instead of an `ostream` object and adjust explanations accordingly.

[P0053R1](#) revised [P0053R0 C++ Synchronized Buffered Ostream](#)

- Add remark to note that exchanging the stream buffer while the stream is wrapped causes undefined behavior and added a note to warn stream buffer implementers about the lock being held in `emit()`. Call `setstate(badbit)` if IO errors occur in `emit()`.
- Replace code references to `basic_streambuf` by the term *stream buffer* introduced in `[stream.buffer]`.
- Provide an example implementation.
- The lock is to be associate to the underlying `basic_streambuf` instead of the `basic_stream`.
- Added an `Allocator` constructor parameter.
- Moves destructor example to `emit()`.
- Clarifies wording about synchronization and flushing (several times).
- List the new header in corresponding table.
- Provide type aliases in `<iosfwd>`.
- Removed copy constructor in favor of providing `get()`.
- Notify that move construction and assignment is deleted.
- Moved class `noteflush_streambuf` into an implementation note.
- Add a design subsection that states that a header test is a sufficient feature test.

[P0053R0](#) revised [N4187 C++ Ostream Buffers](#)

- Updated introduction with recent history.
- Rename `ostream_buffer` to `osyncstream` to reflect its appearance is more like a stream than like a buffer.
- Add an example of using `osyncstream` as a temporary object.
- Add an example of a `osyncstream` constructed with another `osyncstream`.
- Clarify the behavior of nested `osyncstream` executions.
- Clarify the behavior of exceptions occurring with the `osyncstream` destructor.
- Clarify the deferral of flush from the `osyncstream`'s `streambuf` to the final `basic_ostream`.
- Limit the number of references to `noteflush_stringbuf` in anticipation of the committee removing it from the specification.
- Rename `noteflush_stringbuf` to `noteflush_streambuf` to hide possible implementation details.
- Change the base class of `noteflush_streambuf` from `basic_stringbuf` to `basic_streambuf`.

[N4187](#) revised [N4069 C++ Ostream Buffers](#)

- Added note to sync as suggested by BSI via email.

[N4069](#) revised [N3978 C++ Ostream Buffers](#)

- Added a Design section.
- Clarify the reference capturing behavior of the `ostream_buffer` constructors.
- Added `noexcept` and `const` as appropriate to members.
- Added note on throwing wrapped streams.
- Change the `noteflush_stringbuf` public member variable `needsflush` to a public member query function `flushed`.
- Removed the public member function `noteflush_stringbuf::clear`.
- Minor synopsis formatting changes.
- Incorporated feedback from SG1 and Dietmar Kühl in specific in Rapperswil.

[N3978](#) revised [N3892 C++ Ostream Buffers](#)

- Flush the `ostream` if and only if the `ostream_buffer` was flushed.
- Add the `clear_for_reuse` function.
- Change the design from inheriting from `basic_ostream` to using a `noteflush_stringbuf`, which is a slightly modified `basic_stringbuf`. The modification is to note the flush rather than act upon it.

[N3892](#) revised [N3750 C++ Ostream Buffers](#)

- Change name to `basic_ostream_buffer` and add the usual typedefs.
- Change interface to inherit from `basic_ostream` rather than provide access to a member of that type.
- Add a Revisions section.

References

[PSL]

The Open Group Base Specifications Issue 6, IEEE Std 1003.1, 2004 Edition, functions, flockfile,
<http://pubs.opengroup.org/onlinepubs/009695399/functions/flockfile.html>